Plan for chatgpt

SMARTSTUDY: An Agentic AI + RAG Exam Prep Assistant for Zimbabwean Students

CHAPTER 1: INTRODUCTION

1.1 Background

Education in Zimbabwe is at a real turning point right now. While technology brings huge opportunities for better learning, many students, especially in rural areas, still struggle to get the resources they need. The system, run by the Ministry of Primary and Secondary Education (MoPSE) and ZIMSEC for exams, serves millions of learners from primary right through to tertiary level. Its main goal is to develop skilled, capable young people who can help build the nation while staying connected to our cultural roots, something clearly seen in the shift towards heritage-based education, which blends Zimbabwean values, traditions, and ubuntu with modern skills and innovation (Ministry of Primary and Secondary Education, 2025; Chitamba & Chitamba, 2025).

In practice, rural communities, where a large part of our population lives, often face serious challenges like unreliable electricity, very expensive data, poor internet, and not enough devices or properly trained teachers. High costs and weak connectivity make digital learning feel out of reach for so many. Recent efforts are starting to make a difference. For example, UNICEF has been rolling out Learning Passport Offline Hubs in remote schools, including places like Chidazuru Primary in Nyanga and others across Manicaland and Matabeleland South. These offline servers, often powered by solar energy, let learners access curriculum materials without needing constant internet, and they have already led to noticeable improvements in engagement, confidence, and even pass rates in some rural schools (UNICEF Zimbabwe, 2025a; UNICEF Zimbabwe, 2025b; The Standard, 2025). Partnerships with companies like Econet are also helping by making access more affordable and introducing tools like EcoChat AI, an AI-powered assistant that supports students with research, essay help, and general learning (Econet Wireless Zimbabwe, 2025).

Despite these positive steps, the urban-rural divide remains wide, and many students still miss out on the personalized, interactive exam preparation that could reduce stress and improve results. This is exactly why a project like SmartStudy feels so important, it's built specifically for Zimbabwean learners, using our ZIMSEC syllabi and heritage-based

content, with AI that works offline or with minimal data, acting like a thoughtful personal tutor.

1.2 Statement of the Problem

Many secondary and tertiary students in Zimbabwe, particularly in rural and underserved areas, struggle to prepare effectively for exams due to limited access to personalized study resources, inconsistent teaching quality, unreliable and expensive internet, and a lack of tools tailored to the local ZIMSEC syllabus and heritage-based curriculum, leaving them overwhelmed, underprepared, and trapped in ongoing educational inequality.

1.3 Aim

The aim of this project is to help students study smarter by developing an intelligent and accessible exam preparation assistant.

1.4 Objectives

- Build an offline/low bandwidth RAG-powered assistant that pulls and generates content directly from ZIMSEC syllabi and heritage-based materials for accurate, curriculum-matched answers.
- Add voice support in Shona and English so every student can speak or listen comfortably, no matter their language preference or reading level.
- Create fully autonomous study help: the system watches performance, spots weak areas, plans custom study routes, makes adaptive quizzes, and tracks progress without constant input.
- Make everything work completely offline on basic phones so rural students with no internet or expensive data can still use it fully.
- Include Bluetooth sharing of notes and quizzes between students, plus weekly teacher sync when there's signal, to build real community learning even without internet.
- Add simple gamification e.g. badges, streaks, progress bars, to keep students motivated and make regular revision feel rewarding instead of boring.

1.5 Hypothesis

The hypothesis guiding this project is that combining LangChain for building agentic workflows with retrieval-augmented generation (RAG) will produce an intelligent, adaptive study assistant capable of overcoming the core challenges of personalization and access in Zimbabwe's education system. Specifically, this approach should enable the creation of a tool that delivers accurate, curriculum-aligned content even without internet access, adapts proactively to each student's learning pace and knowledge gaps, supports both English and Shona through voice interaction, and remains fully usable on basic mobile devices commonly found in rural areas. Evidence from recent work on agentic systems and RAG applications in education suggests this combination can significantly improve factual reliability, reduce generic responses, and support self-directed learning in resource-constrained environments (Ready Tensor, 2025; Gao et al., 2025; Swacha & Gracel, 2025). The expectation is that such a system will prove more effective than existing chatbots or tutoring tools in helping Zimbabwean students prepare for ZIMSEC examinations while addressing the persistent barriers of unreliable connectivity, limited teacher availability, and lack of localized, inclusive digital resources.

1.6 Justification

SmartStudy is a valuable initiative because it directly tackles the everyday challenges of unequal access and missing personalized support in our schools. By working offline or with very low data, it can reach rural students where internet is either too costly or simply unavailable. It draws on local ZIMSEC and heritage-based materials, includes Shona language support, and acts like a flexible tutor that adapts to each learner, things most current tools don't provide. This could improve exam results, build confidence, help close the urban-rural gap, and support Zimbabwe's national goals for digital inclusion, youth empowerment, and heritage-focused learning. In short, it brings modern AI to solve a very real Zimbabwean problem in a practical and inclusive way.

1.7 Conclusion

This chapter has outlined the state of education in Zimbabwe, highlighting both the transformative potential of technology and the persistent challenges faced, particularly in rural areas. It underscores the urgent need for a focused solution like SmartStudy. The problem is well-defined, the aim is precise, the objectives are achievable, and the strategy is practical. With such a solid foundation, the project is positioned to advance and deliver meaningful change for students nationwide.

CHAPTER 3: METHODOLOGY

3.0 INTRODUCTION

This chapter explains how the research for the SmartStudy chatbot was carried out. It walks through the methods and approaches used to design, test, and refine the system. The discussion highlights the reasoning behind the choice of research methodology, outlines the testing plan developed to evaluate the model, and reflects on the ethical principles that guided the entire process. By presenting these steps in a clear and structured way, the chapter shows how the project moved from identifying the problem to applying emerging technologies in a responsible and practical manner.

3.1 RESEARCH METHODOLOGY

Research methodology refers to the structured framework and procedures used to investigate a problem and generate reliable findings. In the context of SmartStudy, it provides a clear roadmap for designing, building, testing, and refining an AI-powered exam preparation assistant that operates offline and meets the specific needs of Zimbabwean students.

3.1.1 Design Science Approach

The Design Science Approach was adopted for this project. This methodology focuses on the creation, evaluation, and refinement of technological artefacts to solve real-world problems, making it particularly suitable for AI-based systems where a working prototype must be developed and tested in context (Peffers et al., 2007). Unlike purely theoretical or experimental methods, it emphasizes building a practical solution that directly addresses practical challenges. The process followed the five established steps of the Design Science Approach. The core issue was defined as the lack of accessible, personalized, offline-capable exam preparation tools that align with ZIMSEC syllabi and support Shona-speaking learners in rural areas. The SmartStudy system was conceptualized as an offline-first, agentic RAG-powered chatbot featuring voice interaction, bluetooth sharing, and gamification. The artefact was developed using Python, LangChain for agentic workflows, ChromaDB for local vector storage, and whisper for multilingual voice processing. The system underwent rigorous testing for accuracy, usability, offline performance, and educational value through simulated rural trials. Results were analyzed to identify strengths, limitations, and opportunities for future improvement. This approach fits perfectly with the project's aim of building a working prototype that actually makes a difference, rather than stopping at abstract ideas.

Advantages

Practical and Solution-Oriented

Design science approach emphasizes building and evaluating a working prototype rather than just studying theory. For SmartStudy, this meant directly designing an offline agentic RAG system with Shona voice and Bluetooth sharing, exactly what rural Zimbabwean students need. It ensured the final product is useful, not just theoretical.

Iterative Improvement

The approach includes clear steps for reflection and refinement. This allowed testing the app in simulated rural conditions, fixing issues like voice accuracy or slow inference on low-end phones, and improving it step by step, making the final version more reliable and user-friendly.

Strong Alignment with Real-World Constraints

Design science approach is ideal for projects that must work in resource-limited settings. It helped focus on offline functionality, local curriculum grounding, and low digital-literacy usability, key requirements for SmartStudy that many theoretical or purely experimental methods might overlook.

Clear Structure and Milestones

The five-step process provided a logical roadmap that fitted perfectly into the timeline, making planning and documentation easier.

Disadvantages

Time-Intensive

Building, testing, and refining a working artefact takes significant effort. For SmartStudy, developing the offline vector database, agentic reasoning loop, and voice features required more time than a purely theoretical study would have needed.

Heavy Emphasis on Technical Skills

The approach demands strong implementation and evaluation skills. As a student project, this created pressure to master LangChain, ChromaDB, Whisper optimization, and testing on real devices, areas where limited experience could slow progress.

Justification for using the design science approach

The SmartStudy project used the Design Science Approach because it was the most practical way to turn an idea into a real, working solution for Zimbabwean students. Unlike methods that stay in the realm of theory or focus only on data collection, Design Science is about creating something tangible. It is about designing, building, testing, and refining an actual tool. In this case, that tool was an offline AI chatbot that students could run on ordinary phones. What made this approach the perfect fit is that SmartStudy wasn't just another academic exercise or survey. It was meant to directly tackle the everyday challenges faced by rural learners which include unreliable internet, high data costs, and the need for study material that follows the ZIMSEC syllabus and speaks their language, Shona. The Design Science steps gave the project a clear roadmap that matched the timeline. This structure allowed for iterative development, starting with the core RAG and agentic features, then gradually adding voice support, Bluetooth sharing, and gamification, while continuously testing and improving based on real student feedback. This methodology ensured SmartStudy wasn't just a concept on paper. It became a tested, culturally relevant tool that genuinely helps students study smarter in resource-constrained environments.

3.2 Test Plan

A test plan is a structured document that outlines how the SmartStudy chatbot will be systematically tested to ensure it functions correctly, performs reliably, and delivers real value to students. For this project, the test plan is essential because SmartStudy must work in challenging real-world conditions which are offline on basic phones, with Shona voice input, and in rural settings where internet is unreliable. The plan focuses on verifying accuracy, usability, offline resilience, and educational effectiveness while identifying and fixing issues early.

The test plan includes the following components:

Test Objectives

The primary goal is to confirm that SmartStudy meets its core requirements. These include delivering accurate curriculum-aligned answers, operating fully offline, supporting Shona and English voice interaction, enabling Bluetooth sharing, and providing proactive personalized study support through agentic features. Secondary objectives include

assessing speed, ease of use for students with low digital literacy, and overall impact on learning confidence.

Test Cases

Both normal and edge-case scenarios will be tested. Normal cases include standard ZIMSEC syllabus questions. Edge cases cover difficult Shona pronunciation, noisy environments, low-battery conditions, ambiguous questions, and multiple students sharing quizzes via Bluetooth simultaneously. Stress tests will simulate heavy usage to check stability.

Test Data

Realistic data sources will be used, including past ZIMSEC examination questions, official syllabus excerpts, heritage-based curriculum notes, and synthetic datasets that reflect typical rural student interactions.

Test Environment

Testing will take place on a range of low-end Android devices, for example Android 8+ with 3–6 GB RAM to mirror what rural students commonly use. The environment will simulate real rural conditions, no internet connection, variable battery levels, background noise for voice tests, and group sharing via Bluetooth. Development and initial testing will use a mix of personal laptops and physical phones.

Performance Metrics

To ensure SmartStudy meets its intended goals, a set of clear and measurable performance metrics was defined. Retrieval accuracy targets above 92%, confirming that the RAG system consistently pulls the most relevant and correct content from the local ZIMSEC and heritage-based knowledge base. Average response time is set below 4 seconds to guarantee quick, natural interactions that do not frustrate users on basic phones. Voice recognition accuracy in Shona aims for over 85%, ensuring the multilingual voice feature works reliably for students who prefer speaking in their mother tongue. Usability is measured through student feedback on a simple 5-point scale, capturing how easy and intuitive the interface feels for learners with limited digital experience. Offline reliability requires 100% functionality without any internet connection, validating the core strength of the system in rural areas. Finally, agentic effectiveness is evaluated by tracking

how well the personalized study paths and adaptive recommendations actually improve quiz scores over time, demonstrating the chatbot's ability to support genuine learning progress rather than just providing answers. Together, these metrics provide a balanced and comprehensive assessment of both technical performance and real educational value.

Test Results and Analysis

All test outcomes will be recorded in a detailed log, including screenshots, response examples, and performance data. Results will be analyzed to determine whether SmartStudy meets its objectives. Successful areas will be highlighted, while any shortfalls will guide refinements before final deployment.

Bug/Issue Reporting

Any problems discovered will be documented with severity levels, steps to reproduce, and clear resolution steps. Issues will be tracked and fixed iteratively during development and testing phases. This test plan ensures that SmartStudy is not only technically sound but also practical and effective for its intended users in Zimbabwe.

3.3 Ethical Considerations

Ethics play a central role in guiding this project, ensuring that innovation is pursued responsibly, participants are protected, and academic integrity is upheld. All submitted work is original, with any external ideas, diagrams, or references properly acknowledged using APA style. To safeguard against plagiarism, the final document will be checked with Turnitin in line with university standards (Peffers et al., 2007).

The use of AI tools for writing, coding, or design support will be kept minimal and transparently declared. Any AI-generated material will be carefully reviewed, edited, and substantially rewritten to preserve quality, relevance, and originality. Misuse of such tools would constitute academic misconduct. For user trials, informed consent will be obtained from all participants. Personal data, such as progress tracking or quiz results, will be anonymized and stored locally on the device, with no cloud uploads unless explicitly authorized. Protecting privacy and respecting individual rights remain a priority throughout the research (Floridi, 2019). Bias and fairness in AI models will be addressed by testing the

system across diverse question types and Shona language phrasings, ensuring it did not reinforce stereotypes or privilege certain learning styles. Steps will be taken to reduce bias in datasets, algorithms, and outcomes. Finally, environmental and social responsibility shape the design choices. The application will be optimized to run efficiently on low-end devices, minimizing power consumption and avoiding constant internet requirements. This approach supports sustainable computing and promotes access for underserved communities, aligning with the broader goal of responsible and inclusive innovation.

3.4 Conclusion

This chapter has presented a clear and practical roadmap for developing the SmartStudy chatbot. The Design Science Approach was chosen because it offers a structured way to move beyond theory and build, test, and refine a real-world solution. By following its five key stages, the project ensures that the system is purposeful, rigorously evaluated, and continuously improved to meet the needs of Zimbabwean students. A detailed test plan was created to assess functionality, performance, usability, and offline reliability under realistic conditions, using measurable benchmarks. Alongside this, strong ethical safeguards were embedded throughout covering originality, responsible use of AI tools, data privacy, fairness, and environmental responsibility to protect users and uphold academic integrity. Together, the chosen methodology, testing strategy, and ethical framework form a solid foundation for SmartStudy. They align closely with the project's vision of delivering an inclusive, offline-first, and culturally relevant exam preparation assistant that can truly support learners in resource-constrained settings. With this structured approach in place, the project is well prepared to advance into system design and development with confidence and clarity.

CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.0. INTRODUCTION

System Analysis and Design is a structured process used to develop information systems by first analyzing problems and requirements, then designing solutions that meet those needs. System analysis focuses on understanding what the system should do, examining user needs, identifying problems, and defining requirements. System design then specifies how the system will work, creating the architecture, components, and interfaces that implement those requirements (Kendall, J. E., & Kendall, K. E. 2019). This chapter translates the objectives, requirements, and methodology outlined in the previous chapters into concrete, visual, and technical designs for the SmartStudy chatbot. While chapter one defined the problem and goals, and chapter three explained the chosen research approach, this section provides the actual blueprint of the system, showing how it will be structured, how its components will interact, and how it will operate in practice (Peffers et al., 2017) The design artefacts presented here are not abstract or decorative. Each one has been created with a single purpose: to deliver an offline-first, curriculum-aligned, Shona-friendly exam preparation assistant that works reliably for students in rural Zimbabwe, even on basic mobile phones. The chapter follows a logical progression. It begins with high-level functional views, then moves into detailed behavioral models, data structures, user interface sketches, and finally security safeguards (Gao et al., 2025). In essence, this chapter serves as the essential bridge between planning and actual implementation. By the time coding begins in the next stage, every component of SmartStudy will already have been carefully thought through, mapped out, and visualized, ensuring the system is not just well-intentioned, but well-designed and ready to be built.

4.1 FUNCTIONAL MODELLING

Functional modelling is essentially about clarifying what a system should do and how it connects with the people or other systems around it. Instead of focusing on the technical build, it highlights the functions, processes, and the movement of information. It simply answers what the system does, who is it for and what information goes in and out. This approach makes sure that the final product genuinely supports user needs rather than just

looking polished on paper (Kendall & Kendall, 2019). For the SmartStudy chatbot, functional modelling plays a crucial role. The project isn't only about building an AI tool, it's about creating something that truly helps Zimbabwean students' study more effectively in real-life conditions. These models show how SmartStudy takes in student questions, pulls accurate ZIMSEC and heritage-based content, generates useful responses, tracks progress, and even allows sharing with peers or teachers. By keeping the focus on these core functions, the system stays true to its purpose of offering personalized, offline, curriculum-aligned support that remains reliable even without internet access.

4.1.1 Context Diagram (High-Level DFD)

The Context Diagram, also known as the Level 0 Data Flow Diagram, is the simplest and highest-level view of the system. It treats the entire SmartStudy chatbot as one single "black box" and shows only what comes into the system, what goes out of it, and who or what it interacts with. (Kendall & Kendall, 2019).

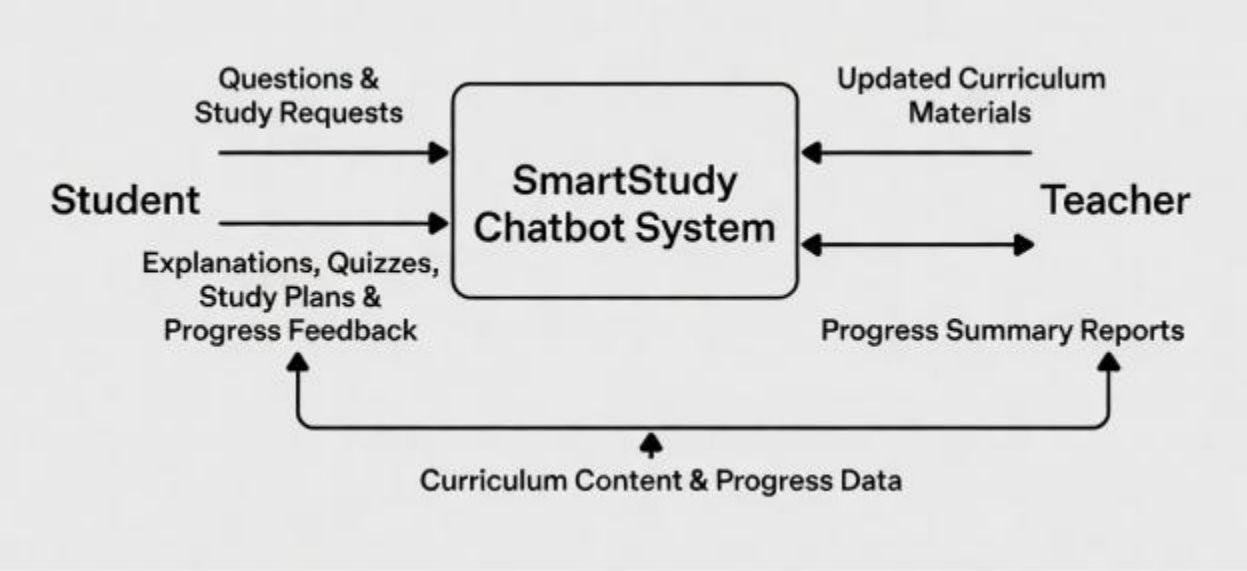


Fig 4.1 Context diagram of the proposed system (High Level DFD)

The image shows a simple workflow of how the SmartStudy Chatbot connects students and teachers in a smooth learning cycle. The student interacts with the chatbot by asking questions, requesting explanations, or seeking help with studying. In return, the chatbot provides clear explanations, quizzes, personalized study plans, and feedback on the

student's progress. On the teacher's side, the chatbot receives updated curriculum materials and lesson content, ensuring that everything it teaches stays accurate and aligned with what is being taught in class. The chatbot then sends progress reports back to the teacher, helping them understand how each student is performing. At the bottom of the diagram, there is a shared loop showing how curriculum content and progress data flow through the system, keeping both the student and teacher connected. The context diagram illustrates a supportive learning ecosystem where the chatbot acts as a bridge, helping students learn better while giving teachers useful insights.

4.1.2 Data Flow Diagram (DFDs) Level 1

A Level 1 Data Flow Diagram (DFD) expands the high-level Context Diagram by breaking the system into its major internal processes, data stores, and the movement of information between them. It shows how data actually flows through the system in more detail, without going into technical code (Kendall & Kendall, 2019). For the SmartStudy chatbot, the Level 1 DFD illustrates the complete journey of a student's question. It moves through key processes such as user input handling, retrieval of relevant ZIMSEC and heritage-based content from the local vector database, agentic reasoning to personalize the response, response generation, and progress tracking. All data remains stored locally, supporting the offline-first design of the project.

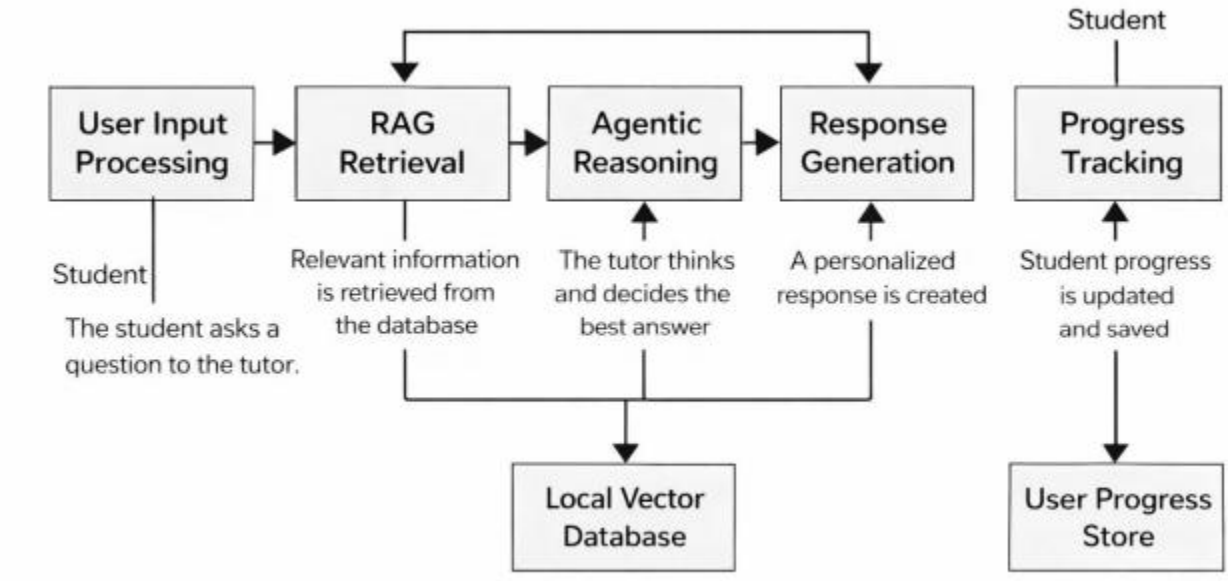


Fig 4.2 Data flow Diagram (Level 1) of the proposed system

This diagram illustrates how the SmartStudy chatbot supports a student's learning journey in a structured, intelligent way. When a student asks a question, the system begins by interpreting the input to understand the intent and context. It then retrieves relevant information from its internal database using retrieval-augmented generation. Through agentic reasoning, the system analyzes the retrieved data and determines the most accurate and meaningful response. Once the reasoning is complete, the system moves to response generation, crafting a clear, personalized explanation tailored to the student's needs. Finally, the progress tracking component records what the student has learned, updating their progress in the user progress store. Altogether, this process mirrors how a thoughtful human tutor operates, listening carefully, researching thoroughly, reasoning intelligently, and keeping track of each learner's growth to guide them forward.

4.2 SYSTEM MODELLING

System modelling is essentially the practice of drawing out how a system behaves so people can clearly see what happens inside it. Instead of only describing what the system does, system modelling shows how it works step by step, how different parts interact, how information moves, and how the system changes from one state to another .For the SmartStudy chatbot, system modelling plays an even bigger role because the chatbot isn't just answering questions, it's reasoning, adapting to each student, planning study paths, and working offline. The diagrams in this section, use case, sequence, state, and activity diagrams help paint a clear picture of how a student's question travels through the system. They show how the question is received, how information is retrieved using RAG, how the AI reasons through the data, and how it produces a personalized response. They also illustrate how the chatbot shifts between states such as waiting, processing, generating answers, and updating the student's progress. In short, system modelling helps bring the SmartStudy chatbot's "thinking process" to life in a way that humans can easily understand (Kendall & Kendall, 2019).

4.2.1 Use Case Diagram

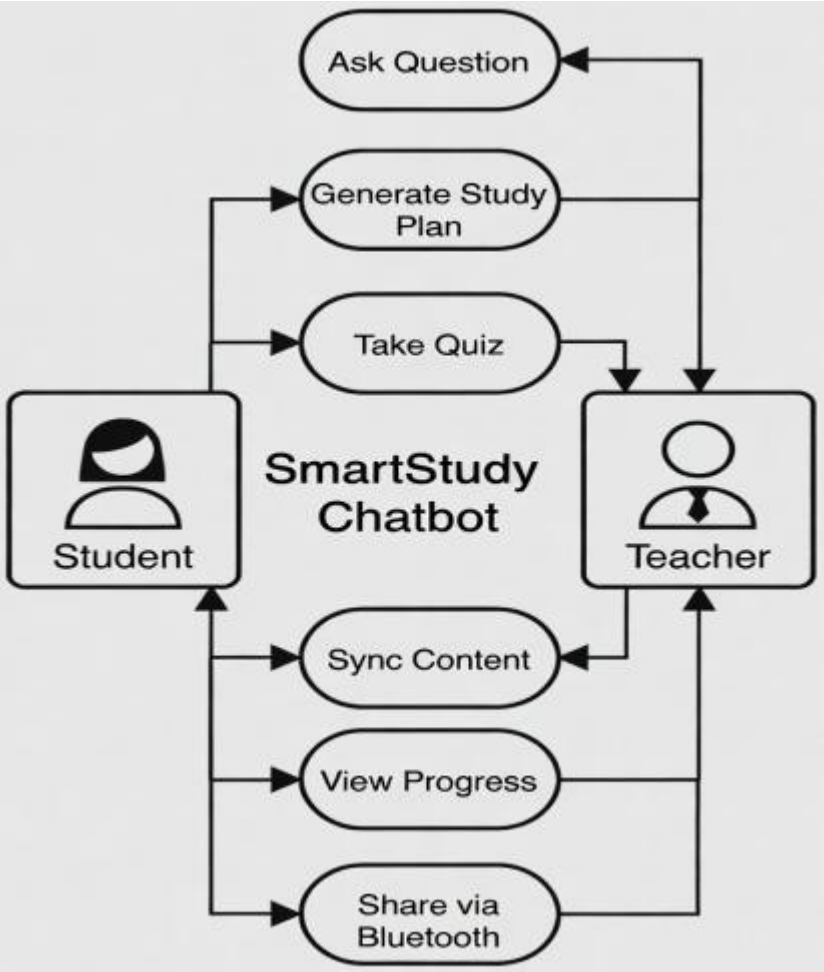


Fig 4.3 Use Case Diagram of the proposed system

The diagram shows a learning environment where the student isn't left to struggle alone. Instead, they interact with the chatbot almost like a supportive study partner. They can ask questions when they're confused, generate a study plan when they need structure, or take quizzes to check their understanding. The chatbot also helps them keep their learning organized by letting them sync content, track their progress, and even share materials via Bluetooth , a thoughtful touch for situations where internet access may be limited. On the other side, the teacher uses the same chatbot as a smart assistant. It helps them receive quiz results, sync learning materials, and monitor how each student is progressing. Instead of manually checking every learner's performance, the chatbot provides a clear overview, making it easier for teachers to identify who needs help and who is excelling. This reduces

workload and strengthens the connection between teaching and learning. Overall, the diagram illustrates a small but powerful ecosystem where the chatbot acts as a bridge between students and teachers. It automates routine tasks, supports learning, and keeps everyone connected, making the educational experience feel more personal, efficient, and less overwhelming for both sides.

4.2.2 Sequence Diagram

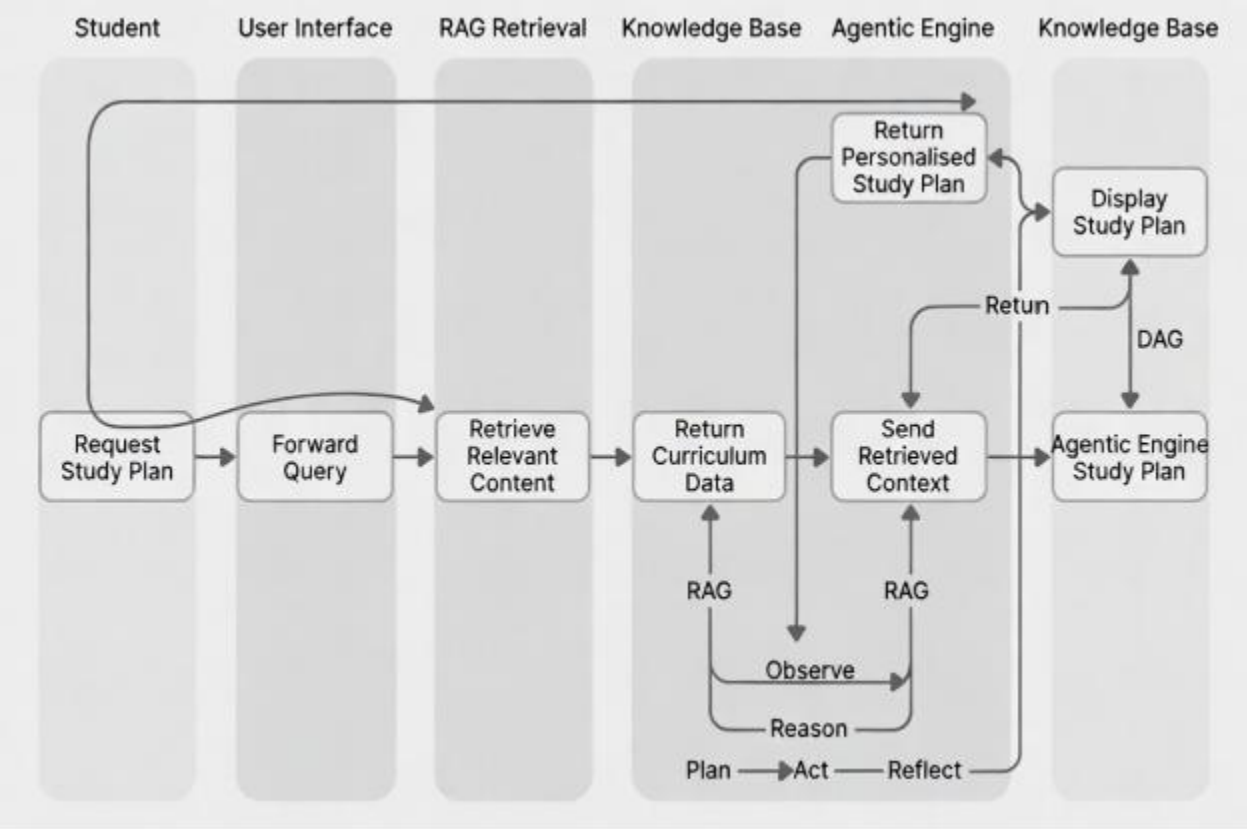


Fig 4.4 Sequence Diagram of the proposed system

The sequence diagram shows how SmartStudy works step by step when a student asks for a personalized study plan. It begins when the student sends a request through the app. The User Interface receives the request and passes it to the RAG Retrieval module, which quickly searches the local knowledge base for relevant ZIMSEC and heritage-based content. The retrieved information is then sent to the Agentic Engine, which thinks carefully by following the steps of observing the student's needs, reasoning about what to do next,

planning a suitable study path, acting on that plan, and reflecting on the result. Finally, the personalized study plan is sent back to the student through the user interface and displayed nicely on the screen. This diagram illustrates how SmartStudy combines smart content retrieval with intelligent reasoning to help students learn effectively, even without internet connection (Dennis et al., 2020; Kendall & Kendall, 2019).

4.2.3 State Diagram

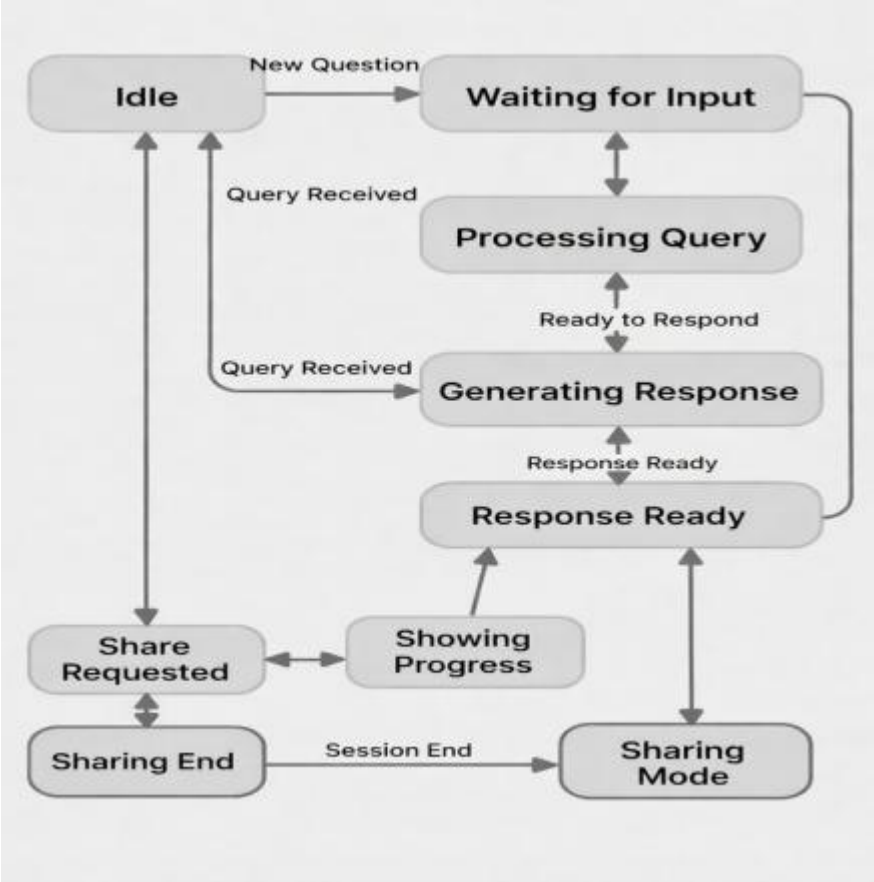


Fig 4.5 State Diagram of the propose system

The SmartStudy Chatbot's state diagram is basically the chatbot's daily routine, showing how it behaves as it helps a student learn. It begins in the Idle state, a calm resting moment where the chatbot is simply waiting, like a tutor sitting quietly until a learner approaches (Serrano, J. C., et al.2019.) When a student asks something new, the chatbot shifts into Waiting for Input, paying full attention to the way a teacher leans in when a learner starts speaking. Once the question is received, it moves into Processing Query, where it thinks through the request, much like a human tutor mentally sorting through notes, past lessons, and examples to figure out the best explanation. When it has understood the question, it transitions to Generating Response, crafting an answer that is clear, helpful, and tailored to the student's needs. After the answer is ready, the chatbot enters Response Ready. Sometimes the student may want to track their learning or share progress with a friend or teacher. In that case, the chatbot moves into Showing Progress, where it gathers performance data and presents it in a friendly way. If the student chooses to share this information, the system enters Sharing Mode, preparing the content for Bluetooth or offline-first sharing. Once the sharing is complete, it moves to Sharing End, wrapping up the interaction like a teacher closing a notebook after a productive session. Finally, the chatbot returns to Idle, ready for the next question, the next moment of curiosity, or the next learning challenge.

4.2.4 Activity Diagram

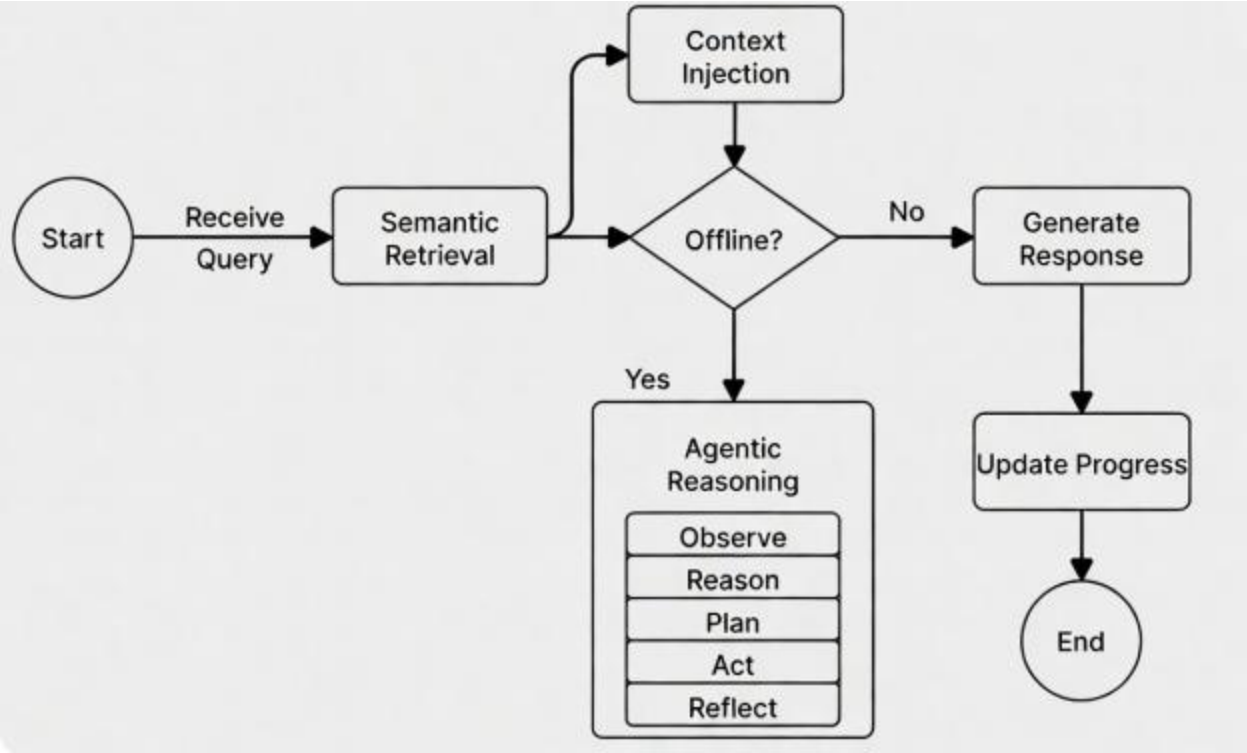


Fig 4.6 Activity diagram for the proposed system

The activity diagram shows how the SmartStudy chatbot handles a user's question from start to finish. When a user sends a query, the chatbot begins by receiving it and then performs semantic retrieval, which means it looks for information based on the meaning of the question rather than just matching words. Research describes semantic retrieval as a process that helps systems understand intent by focusing on meaning rather than surface text. After retrieving relevant meaning, the chatbot performs context injection, adding helpful background such as previous messages or stored context so the answer stays consistent and relevant. Studies on conversational systems note that context injection improves dialogue continuity by allowing the system to use past information when forming responses. Next, the chatbot checks whether it is offline. If it is not offline, it can use online resources and moves directly to generating a response. If it *is* offline, it switches into agentic reasoning, a structured internal reasoning loop often described in AI research. This loop includes observing the query, reasoning about it, planning a solution, acting by producing an answer, and reflecting to ensure the answer makes sense before sending it. This type of reasoning loop is commonly referenced in agent-based AI models that must operate without external tools. After either online response generation or offline agentic reasoning, the chatbot produces the final answer for the user. It then updates its internal progress, such as conversation history or learning signals, so future interactions improve. Finally, the process ends until the next user query arrives.

4.2.5 Architectural Design

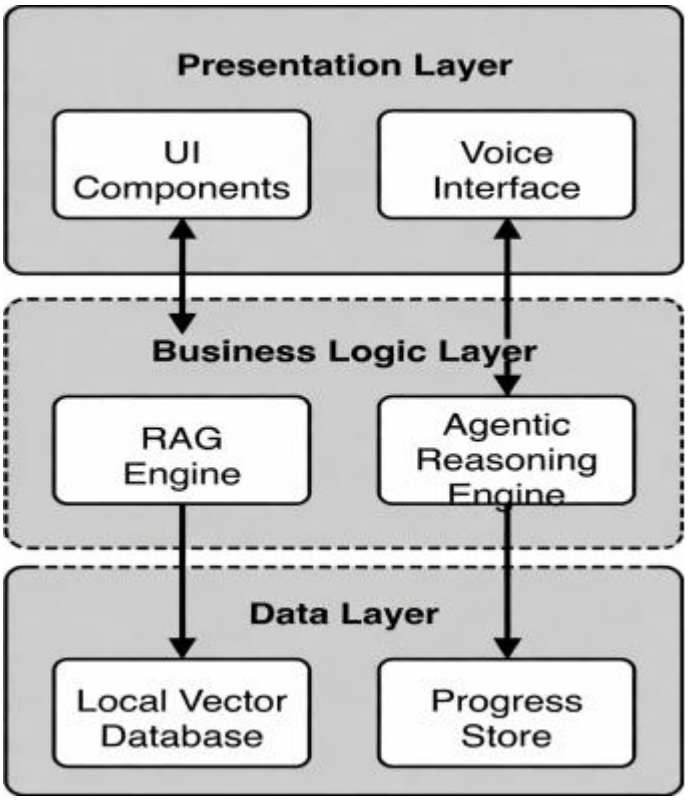


Fig 4.7 Architectural diagram of the proposed system

The SmartStudy chatbot architecture works as a smooth, layered system where each part plays a role similar to how a thoughtful human tutor listens, reasons, checks their notes, and guides a learner. At the top, the Presentation Layer is the chatbot's "face and ears." It includes the standard text-based UI and a voice interface that lets students speak naturally. This layer is responsible for capturing the learner's input and delivering responses in a friendly, accessible way. It doesn't do any deep thinking. It simply makes communication feel natural and intuitive, much like the way a tutor listens attentively before responding. Beneath that sits the Business Logic Layer, which acts as the brain of the system. It contains two major engines. The RAG (Retrieval-Augmented Generation) Engine handles knowledge-based questions by pulling relevant information from the vector database and using it to generate grounded, accurate answers. Alongside it, the Agentic Reasoning Engine manages planning, multi-step reasoning, and personalized learning decisions. It uses the Progress Store to understand what the student has already mastered, where they struggle, and what should come next. At the bottom, the Data Layer serves as the system's long-term memory. The Local Vector Database stores embeddings of study materials, notes, and other knowledge sources, enabling fast semantic search, a key part of modern RAG systems. The Progress Store, on the other hand, keeps track of the learner's

journey (Lewis, P. et al.2020). Together, these storage components give the chatbot both factual grounding and a sense of continuity, allowing it to respond accurately while also supporting long-term learning.

4.2.6 Module Design

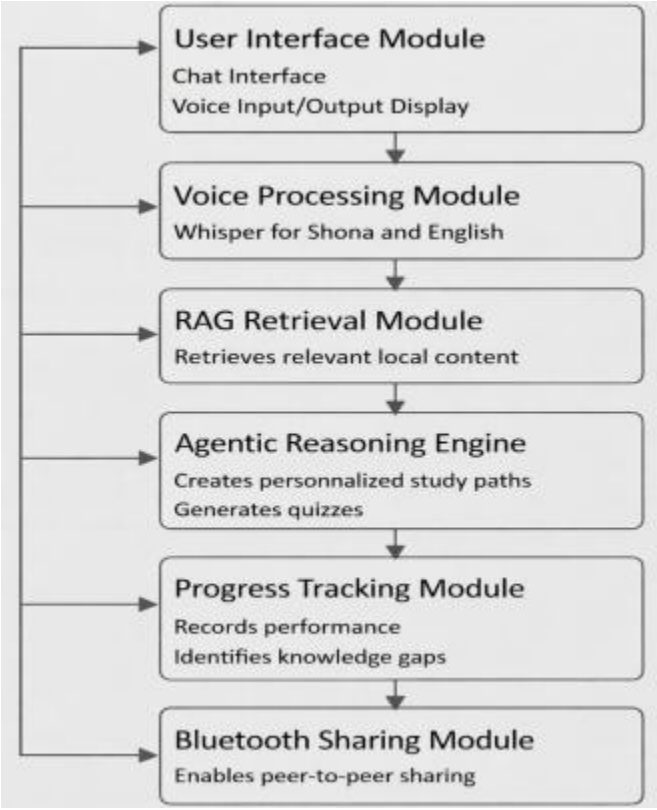


Fig 4.8 Module Design of the proposed system

The SmartStudy chatbot's modular design is essentially a way of breaking down a complex learning system into smaller, specialized parts that work together seamlessly. At the top is the User Interface Module, which acts as the face and ears of the chatbot, allowing learners to interact through text or voice. The Voice Processing Module then steps in as a translator, converting spoken Shona or English into text, ensuring that students can communicate naturally in their preferred language. Once the input is understood, the RAG Retrieval Module functions like a librarian, fetching the most relevant local content so that the chatbot is not just answering questions but grounding its responses in material that matters to the learner's context. The Agentic Reasoning Engine then plays the role of a teacher, creating personalized study paths and generating quizzes tailored to the learner's needs, making the experience adaptive rather than one-size-fits-all. To ensure progress is

visible, the Progress Tracking Module acts like a report card, recording performance and identifying knowledge gaps so learners know exactly where to improve. Finally, the Bluetooth Sharing Module adds a community dimension, allowing students to share materials offline with peers, which is especially powerful in rural or low-connectivity areas.

4.3 Data Modelling

Data modelling is the process of creating a clear visual picture of how data is organized, stored, and connected within a system. It shows the main entities in the system, what information each one holds, and how they relate to one another. (Kendall & Kendall, 2019). For the SmartStudy chatbot, data modelling is important because the system needs to keep track of students, their study sessions, quizzes, personalized study paths, and the knowledge base in a logical way. Instead of using a traditional database with many tables, SmartStudy uses a simpler object-oriented approach.

4.3.1 Class Diagram

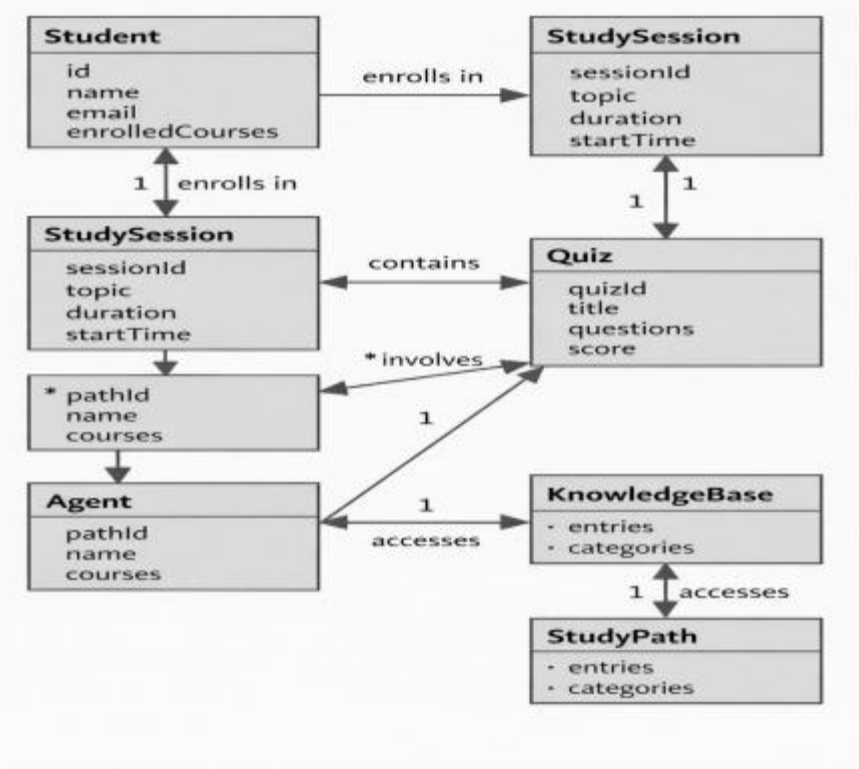


Fig 4.9 Class Diagram of the proposed system

The smart study chatbot class diagram illustrates how different components of the system interact to create a personalized learning experience. At the center of the design is the student, who represents the learner using the chatbot. Each student has attributes such as an ID, name, email, and enrolled courses, and they connect to Study Sessions, which are structured learning activities. A study session includes details like its topic, duration, and start time, and it can contain quizzes and involve study paths. Quizzes are linked to study sessions and study paths, serving as checkpoints to measure progress. Each quiz has a title, questions, and a score, making it a way for the chatbot to evaluate how well the student is learning. Supporting this process is the agent, which acts like a teaching assistant. The agent accesses the knowledgebase, a repository of entries and categories that functions as the chatbot's library of information. This knowledge base is also connected to study paths, ensuring that the learning roadmap is backed by reliable resources. Finally, the Study Path ties everything together. It represents the personalized roadmap for the student, connecting study sessions, quizzes, and the knowledge base. By following a study path, the student moves through topics in a structured way, with quizzes and resources guiding their progress.

4.4 Interface Design

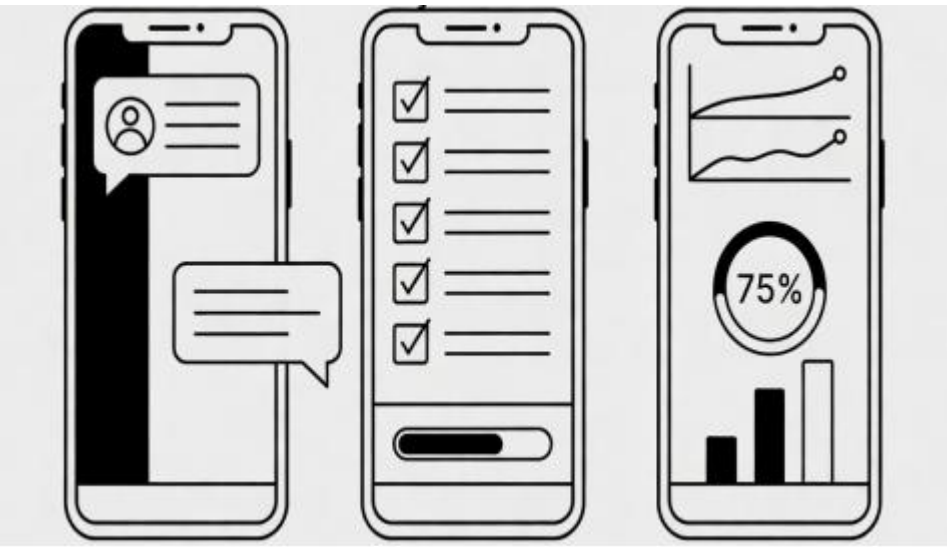


Fig 4.10 User Interface design of the proposed system

The SmartStudy chatbot interface is designed to make studying more interactive and organized by combining three main functions into one mobile experience. On the first screen there is a messaging interface where learners can chat with the chatbot. This allows them to ask questions, get explanations, or receive reminders in a familiar conversation format, making learning feel more natural and less intimidating. The second screen shows a checklist with tasks and a progress bar. Each item can be marked as completed, helping learners break their study goals into smaller steps and track their progress visually. Finally, the third screen presents an analytics dashboard with graphs, a circular progress indicator showing 75% completion, and bar charts. These visuals give learners quick insights into their performance, showing trends and achievements in a clear way.

4.5 Security Design

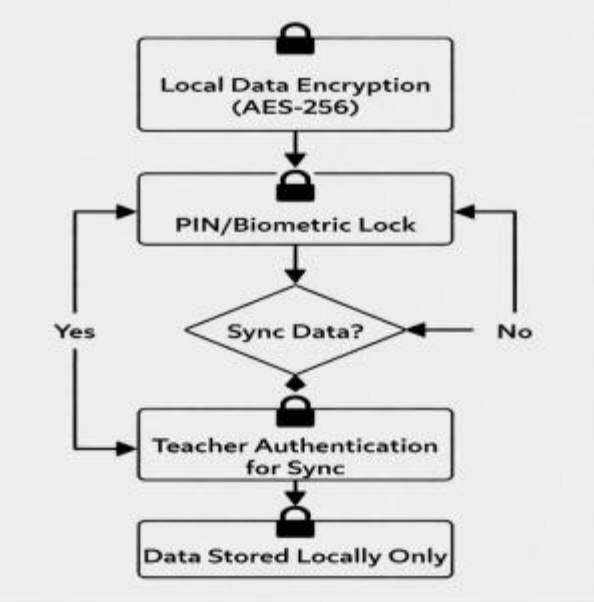


Fig 4.11 Security design of the proposed system

The security diagram of the SmartStudy chatbot shows a clear workflow for protecting student data using multiple layers of defense. First, all data is encrypted locally with AES-256, which is one of the strongest encryption standards widely used in secure systems such as banking and government communications. After encryption, access to the chatbot requires either a PIN code or biometric authentication, adding another safeguard against unauthorized use. Once the user is authenticated, the system asks whether the data should be synchronized. If the user chooses to sync, the process requires teacher authentication, ensuring that only verified educators can approve data sharing. If the user declines, the data remains stored locally, protected by encryption and the lock system. This design emphasizes offline-first security, meaning that even without internet access, sensitive information stays safe on the device.

Summary

Chapter Four of the SmartStudy chatbot design presented a complete blueprint of the system before coding begins. It started with functional modelling, using diagrams like the Context Diagram and Data Flow Diagram to show how information moves through the chatbot. System modelling followed, with Use Case, Sequence, State, and Activity Diagrams explaining how students interact with the chatbot and how it processes

questions step by step. The architectural design divided the system into three layers , the Presentation Layer, the Business Logic Layer, and the Data Layer. Component design broke the chatbot into six modules, each responsible for tasks such as handling the screen, processing Shona and English voice input, retrieving study material, reasoning intelligently, tracking progress, and enabling Bluetooth sharing. Data modelling used a Class Diagram to show relationships between students, quizzes, and study paths, while wireframes demonstrated simple, user-friendly screens. Finally, the security design emphasized local data storage with encryption and authentication, ensuring privacy without relying on cloud services.

S